Wang Ma

Troy, NY • +1 518-961-7579 • maw6@rpi.edu
wangma2002@gmail.com • wma17.github.io

Summary

Ph.D. student in Bayesian deep learning and uncertainty quantification, with industry experience building reliable generative AI systems at Microsoft and scalable uncertainty methods at IBM Research. Seeking Summer 2027 research and applied science internships in trustworthy AI, generative AI, and machine learning systems.

Education

Rensselaer Polytechnic Institute (RPI) Aug 2024 – May 2029 (Expected)
Ph.D. in Computer & System Engineering (Advisor: Prof. Qiang Ji) *Troy, NY*
Research Interests: Uncertainty Quantification, Bayesian Deep Learning, Knowledge-augmented AI

Southern University of Science and Technology (SUSTech) Aug 2020 – July 2024
B.S. in Data Science and Big Data Technology (Advisor: Prof. Chao Wang) *Shenzhen, China*

University of California, Irvine (UCI) Mar 2023 – July 2023
Exchange Student (ASAP Program) *Irvine, CA*

Publications & Manuscripts

1. **Towards Knowledge-augmented Bayesian Deep Learning For Computer Vision**
   Wang Ma, Hanjing Wang, Yufei Zhang, Darsha Udayanga, Qiang Ji.
   *CVPR 2026. **Highlight**.*
2. **Black-Box Uncertainty Quantification for Large Language Models via Ensemble-of-Ensembles**
   Wang Ma, Debarun Bhattacharjya, Junkyu Lee, Nhan Pham, Harsha Kokel, Qiang Ji.
   Submitted to NeurIPS 2026. Preliminary version accepted to AAAI 2026 AIR-FM Workshop.
3. **Distilling Bayesian Uncertainty into a Single Forward Pass**
   Wang Ma, Qiang Ji.
   Submitted to NeurIPS 2026.
4. **Causal Saliency Map For Post-hoc Model Explanation**
   Hanjing Wang, **Wang Ma**, Naiyu Yin, Qiang Ji.
   Submitted to NeurIPS 2026.
5. **CURE: Causal Uncertainty correlation REattribution via Interventional Decomposition**
   Wang Ma*, Ali Najibi*, Naiyu Yin, Qiang Ji. (*Equal Contribution)
   Submitted to NeurIPS 2026.

Industrial Experience

Microsoft **Redmond, WA**
Applied Science Intern, PPT ML Team *May 2026 – Aug 2026*

- Improved presentation quality for PowerPoint's **PPT Generation Agent** by diagnosing and resolving table-rendering issues in agent-generated slides.
- Optimized the agent's **build pass** to reduce end-to-end generation latency and streamline the slide-creation workflow.

IBM Research **Yorktown Heights, NY**
Research Extern (Mentor: Dr. Debarun Bhattacharjya) *May 2025 – Aug 2025*

- Developed a **perturbation-based UQ framework** for **black-box LLMs**, decomposing uncertainty without access to model weights or intermediate activations.
- Introduced an "ensemble-of-ensembles" method to disentangle **aleatoric and epistemic uncertainty**; first-authored paper accepted to AAAI 2026 AIR-FM Workshop and extended to NeurIPS 2026 submission.

Academic Experience

Rensselaer Polytechnic Institute (RPI) **Troy, NY**
Graduate Research Assistant (Advisor: Prof. Qiang Ji) *Sept 2024 – Present*

- **Knowledge-Augmented Bayesian Deep Learning:** Proposed a two-stage framework with a **knowledge-induced prior** and **adaptive knowledge likelihood**, improving accuracy, robustness, and **OOD detection**.
- **Causal Saliency Maps:** Implemented **Causal Markov Blanket (CMB)** detection algorithms using label-wise mutual information to identify causally relevant features and filter spurious correlations.
- **Semantic Uncertainty Disentanglement:** Developed a **contrastive learning framework** to disentangle semantic and nuisance uncertainties, achieving strong benchmark performance. (Technical Report)
- **Efficient Single-Model UQ:** Used **NTK theory** and **credal interval-based methods** to study practical uncertainty quantification with a single auxiliary neural network. (Technical Report)

SUSTech **Shenzhen, China**
Undergraduate Researcher (Advisor: Prof. Chao Wang) *Mar 2024 – Aug 2024*

- **Hyperspectral Image Restoration:** Built a self-supervised method combining **Implicit Neural Representations (INR)** and **Diffusion Models**.
- **Unpaired Image Denoising:** Optimized a **VAE** from unpaired data with Mutual Information maximization.

Technical Skills

- **Programming:** Python (PyTorch, NumPy, Pandas, Scikit-learn), Java, MATLAB, LaTeX
- **Developer Tools:** Git, Docker, Linux
- **Natural Languages:** Mandarin Chinese (Native), English (Fluent), Japanese (Conversational)

Honors & Service

- **Golden Reviewer**, International Conference on Machine Learning (ICML) 2026
- **SUSTech Excellent Undergraduate Graduation Project** (Thesis: End-to-end Unpaired Image Denoising) June 2024
- **Seminar Organizer:** AI Optimization & Theory (SUSTech); led sessions on Bayesian Optimization & Bayesian Deep Learning 2024
- Second Prize in Guangdong Province, Mathematical Contest in Modeling 2021
- **Scholarships:** SUSTech Excellent Student Scholarship (2021, 2022)